

Supplementary Revision Addendum

Trustworthy Medication Recommendation: A Safety-Centered Survey of Knowledge-Grounded and LLM-Assisted Methods

Targeted revision update and endpoint recheck: 4 October 2026

R1. Scope and study-count correction

This addendum supersedes the included-study roster and appraisal totals in the earlier Supplementary File S1. It preserves the original search documentation; it does not assert a new exhaustive database search or reconstruct unavailable screening logs. The original manuscript reports 132 identified records, 39 duplicates, 93 screened records, 37 title/abstract exclusions, and 56 retained full texts for the search through 20 May 2026.

The original roster includes MetaCare++ [43], which predicts diagnoses rather than medications. The earlier S1 itself marks its MR appraisal as not applicable. Removing this entry leaves 55 studies carried forward from the original roster. Five absent full-text studies are added through the targeted revision route, producing a 60-study updated qualitative synthesis. This update includes previously missed pre-cutoff papers as well as later publications; publication coverage and original database-search coverage are distinct.

GAMENet [39], SafeDrug [51], and G-BERT [38] remain pre-2022 background references. They do not enter the 60-study register or the 60-study appraisal. The current roster does not count the other eight 2026 references previously added for narrative coverage as screened studies. Family examples and proposed frameworks in the manuscript are illustrative, rather than corpus-frequency statistics.

R2. Targeted full-text additions and contextual records

The following five studies meet the existing patient-specific medication or sequential-regimen selection criterion and have full-text evidence available for this update:

- RPNNet [15]: uploaded publisher PDF, 12 pages; Neural Networks 198 (2026), 108618. Online publication: 21 January 2026.
- Nephrotoxic medication management RL [11]: uploaded publisher PDF, 15 pages; Biomedical Signal Processing and Control 112 (2026), 108577. First available online: 20 August 2025. It is within the original date window despite its 2026 volume year.
- MetaDrug [7]: open author manuscript arXiv:2601.22820v1, accessed 3 October 2026; ICDE 2026 status confirmed by the author’s publication list.
- HypeMed [58]: open 28-page author PDF at <https://arxiv.org/pdf/2603.18459>, accessed 3 October 2026; TOIS DOI verified. The inspected PDF includes eICU, unlike an earlier inspected HTML representation; extraction follows the PDF and identifies the absent eICU DDI evaluation.
- GenRxR [10]: open 9-page author PDF at <https://arxiv.org/pdf/2607.24829>, accessed 3 October 2026; RecSys 2026 program and DOI verified. Extraction includes the licensed expert’s synthetic-data plausibility test.

RxGuard [45] is contextual because the primary repository abstract and publication metadata were accessible but the full text was not retrieved for extraction. The blockchain federated-learning paper [1] is contextual because its mixed classification/recommendation tasks do not provide comparable encounter-level medication-set evaluation. Neither record enters the 60-study register. KEHGCN [60] and FLAME [6] are already in the original roster and are not duplicated. IEEE document 11470461 remains unidentified and is not cited on an assumed identity.

R3. Updated included-study register

Table R1 contains 60 studies. O = carried forward from the original S1 after the scope correction; U = targeted full-text addition. Original-roster metadata are carried forward unless explicitly corrected, rather than presented as newly verified full-text extraction. MoleRec’s venue is corrected to WWW 2023; MedRec and HKRec use final issue years 2023 and 2025, respectively. The endpoint recheck corrected dataset labels for REFINE, context-aware MR (Chen et al.), CausalMed, CIDGMed, EDRMM, DNMDR, LEADER, MedAlign, MedGCN, ACDNet, BiMoRec, KEHGCN, BH3-MedRec, LAMO, MedRec, CAMEr, and MKAF-Net. Author code indicates MIMIC-III for Trans-GAHNet, whereas the earlier S1 lists MIMIC-IV; this paper-level discrepancy needs confirmation. HKRec’s indexed abstract describes MIMIC-III and MIMIC-IV, while its authors’ public implementation describes a MIMIC-IV cohort. In particular, MedAlign does not use MIMIC-CXR in its original evaluation, and the MIMIC-III/IV departmental cohorts of BH3-MedRec do not establish an eICU test. These metadata corrections do not change eligibility. FLAME’s open manuscript reports MIMIC-IV and eICU generalization tests and uses parsed clinical notes. The machine-readable file `updated_study_register.csv` gives a source for each rechecked metadata field. Study IDs remain stable when reference numbers change.

Table R1: Updated qualitative-synthesis register (60 studies).

ID	Study/model	Venue (year)	Route	Dataset metadata
MR001	PharmaLLM [2]	HCIS (2024)	O	Private
MR002	REFINE [3]	NeurIPS (2023)	O	MIMIC-IV + private outpatient cohort
MR003	Confidence calibration (Chen et al.) [4]	KDD (2025)	O	MIMIC-III + MIMIC-IV
MR004	Context-aware MR (Chen et al.) [5]	AAAI (2023)	O	MIMIC-III + MIMIC-IV
MR005	FLAME [6]	NeurIPS (2025)	O	MIMIC-III; MIMIC-IV and eICU generalization tests
MR006	GPSRec [8]	CIKM (2025)	O	MIMIC-III + MIMIC-IV
MR007	KRAM [9]	ESWA (2026)	O	MIMIC-III
MR008	HI-DR [12]	AAAI (2025)	O	MIMIC-III + MIMIC-IV
MR009	VITA [13]	AAAI (2024)	O	MIMIC-III + MIMIC-IV
MR010	StratMed [14]	arXiv (2023)	O	MIMIC-III
MR011	CausalMed [16]	CIKM (2024)	O	MIMIC-III + MIMIC-IV
MR012	Trans-GAHNet [17]	ESWA (2024)	O	MIMIC-III (code; verify paper)
MR013	DGCL [18]	JBH (2023)	O	MIMIC-III
MR014	MR-DTR [19]	WWW (2025)	O	MIMIC-III + MIMIC-IV
MR015	CIDGMed [20]	KBS (2025)	O	MIMIC-III + MIMIC-IV
MR016	TreatRAG [21]	RecSys (2025)	O	MIMIC-IV
MR017	EDRMM [22]	BMC Bioinf. (2025)	O	MIMIC-III
MR018	DNMDR [23]	KBS (2025)	O	MIMIC-III
MR019	IMDR [24]	KDD (2025)	O	MIMIC-III + MIMIC-IV
MR020	TEMPT [25]	TOIS (2025)	O	MIMIC-III + MIMIC-IV + eICU
MR021	LEADER [26]	arXiv (2024)	O	MIMIC-III + MIMIC-IV
MR022	SHAPE [27]	JBHI (2023)	O	MIMIC-III
MR023	DKINet [28]	arXiv (2023)	O	MIMIC-III + MIMIC-IV + eICU
MR024	MedAlign [29]	ACM MM (2025)	O	MIMIC-III + MIMIC-IV
MR025	PAUP [30]	IP&M (2026)	O	MIMIC-III + MIMIC-IV
MR026	MedGCN [31]	JBH (2022)	O	MIMIC-III + NMEDW

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Table R1 continued

ID	Study/model	Venue (year)	Route	Dataset metadata
MR027	ACDNet [32]	JB1 (2024)	O	MIMIC-III + MIMIC-IV
MR028	BiMoRec [33]	ESWA (2025)	O	MIMIC-III + MIMIC-IV
MR029	KindMed [34]	arXiv (2023)	O	MIMIC-III
MR030	EGNet [35]	Appl. Intell. (2025)	O	MIMIC-III
MR031	KATMed [36]	JB1 (2026)	O	MIMIC-III/IV + external eICU
MR032	RAG-CPMF [37]	Pharmacological Research (2025)	O	Public CPM instructions + guideline cases
MR033	TAHDNet [40]	JB1 (2022)	O	MIMIC-III
MR034	CEHMR [41]	AIM (2023)	O	MIMIC-III
MR035	4SDrug [42]	KDD (2022)	O	MIMIC-III
MR036	CAMeR [44]	ICONIP (2023)	O	MIMIC-III + MIMIC-IV
MR037	EXCERF [46]	KBS (2025)	O	MIMIC-III + MIMIC-IV
MR038	ARMR [47]	IJCAI (2025)	O	MIMIC-III + MIMIC-IV
MR039	TAKECare [48]	Inf. Fusion (2025)	O	MIMIC-III
MR040	PROMISE [49]	IP&M (2024)	O	MIMIC-III
MR041	COGNet [50]	WWW (2022)	O	MIMIC-III
MR042	SDRBT [52]	JB1 (2025)	O	MIMIC-III
MR043	MoleRec [53]	WWW (2023)	O	MIMIC-III
MR044	AKA-SafeMed [54]	Inf. Sci. (2024)	O	MIMIC-III
MR045	SSPNet [55]	IJCAI (2025)	O	MIMIC-III + MIMIC-IV
MR046	HKRec [56]	TCSS (2025)	O	MIMIC-III + MIMIC-IV (abstract)
MR047	KEDRec-LM [57]	arXiv (2025)	O	MIMIC-IV
MR048	MedRec [59]	TOIS (2023)	O	TCM herbs + private Medicine symptom sets
MR049	KEHGCN [60]	AAAI (2026)	O	MIMIC-III + two MIMIC-IV subsets
MR050	BH3-MedRec [61]	TIST (2026)	O	MIMIC-III + MIMIC-IV department cohorts
MR051	LAMO [62]	arXiv (2025)	O	MIMIC-III training; MIMIC-IV/eICU tests
MR052	RAREMed [63]	SIGIR (2024)	O	MIMIC-III + MIMIC-IV
MR053	MKAF-Net [64]	Appl. Intell. (2025)	O	MIMIC-III + MIMIC-IV
MR054	TCM-KLLaMA [65]	CBM (2025)	O	Public TCM prescription benchmark
MR055	DAI-Net [66]	JBHI (2024)	O	MIMIC-III + MIMIC-IV + eICU
MR056	RPNet [15]	Neural Networks (2026)	U	MIMIC-III + MIMIC-IV
MR057	Nephrotoxic medication RL [11]	BSPC (online 2025) (2026)	U	MIMIC-IV v3.1; same-hospital later cohort
MR058	MetaDrug [7]	ICDE (2026)	U	MIMIC-III + Kansas AKI (within-dataset training/evaluation)
MR059	HypeMed [58]	TOIS (2026)	U	MIMIC-III + MIMIC-IV + eICU (DDI omitted on eICU)
MR060	GenRxR [10]	RecSys (2026)	U	MIMIC-III + MIMIC-IV

R4. Descriptive appraisal of all included studies

The earlier S1 Table S5 contains 51 rows: 50 numeric ratings and one diagnosis-only N/A row. Excluding its three pre-2022 numeric rows leaves 47 eligible original-roster ratings. The prior scores are retained except for the three full-text corrections documented in R7. Full texts were examined for the 13 remaining included studies and new descriptive scores assigned; thus Table R2 and `appraisal_60.csv` now cover the 60-study register. The separate `appraisal_additions_13.csv` records page-specific evidence for the new ratings, which await author cross-check. The initial LLM-assisted appraisal was checked against original papers by the lead author with a second author cross-check of original review materials; the authors report no scoring disagreements. The new 13-score evidence pass is not an independent duplicate appraisal. Neither prespecification nor validated risk-of-bias status is inferred.

C1: MR task definition; C2: dataset/preprocessing; C3: architecture/training; C4: predictive eval-

uation; C5: safety-aware evaluation; C6: interpretability/evidence/explanations; C7: external validation/robustness/deployment. Ratings are 2 (adequate), 1 (partial), 0 (inadequate/not reported). C5–C7 are broad reporting criteria rather than endpoint-specific counts. In particular, C7 is not a count of fixed-model external validation. A methodological component, such as DDI control without a reported outcome, should not be described as demonstrated clinical safety.

Table R2: Study-level descriptive appraisal (60 studies; new scores await author cross-check).

ID	Study/model	C1	C2	C3	C4	C5	C6	C7	Total
MR001	PharmaLLM	1	2	2	1	1	0	1	8
MR002	REFINE	2	2	2	2	2	1	2	13
MR003	Chen et al.	2	1	2	2	1	0	2	10
MR004	Context-aware MR	2	2	2	2	2	1	1	12
MR005	FLAME	2	2	2	2	2	1	2	13
MR006	GPSRec	2	2	2	2	2	1	2	13
MR007	KRAM	2	2	2	2	1	1	2	12
MR008	HI-DR	2	2	2	2	2	1	1	12
MR009	VITA	2	2	2	2	2	2	2	14
MR010	StratMed	2	2	2	2	2	2	2	14
MR011	CausalMed	2	2	2	2	2	2	1	13
MR012	Trans-GAHNet	2	2	2	2	2	1	1	12
MR013	DGCL	2	2	2	2	2	2	2	14
MR014	MR-DTR	2	2	2	2	2	1	2	13
MR015	CIDGMed	2	2	2	2	2	2	1	13
MR016	TreatRAG	2	2	2	2	1	2	2	13
MR017	EDRMM	2	2	2	2	2	1	1	12
MR018	DNMDR	2	2	2	2	2	2	1	13
MR019	IMDR	2	2	2	2	2	1	2	13
MR020	TEMPT	2	2	2	2	0	0	2	10
MR021	LEADER	2	2	2	2	0	0	1	9
MR022	SHAPE	2	2	2	2	2	1	1	12
MR023	DKINet	2	2	2	2	2	2	2	14
MR024	MedAlign	2	2	2	2	2	1	2	13
MR025	PAUP	2	1	2	2	2	1	2	12
MR026	MedGCN	2	2	2	2	0	1	2	11
MR027	ACDNet	2	2	2	2	2	1	1	12
MR028	BiMoRec	2	2	2	2	2	1	1	12
MR029	KindMed	2	2	2	2	2	1	2	13
MR030	EGNet	2	2	2	2	2	2	2	14
MR031	KATMed	2	1	2	2	2	1	2	12
MR032	RAG-CPMF	2	2	1	2	2	2	1	12
MR033	TAHDNet	2	2	2	2	1	2	1	12
MR034	CEHMR	2	2	2	2	2	2	1	13
MR035	4SDrug	2	2	2	2	2	1	2	13
MR036	CAMeR	2	1	1	2	2	2	2	12
MR037	EXCERF	2	2	2	2	2	2	2	14
MR038	ARMR	2	2	2	2	0	1	1	10
MR039	TAKECare	2	2	2	2	2	1	2	13
MR040	PROMISE	2	2	2	2	2	1	2	13
MR041	COGNet	2	2	2	2	2	1	0	11
MR042	SDRBT	2	2	2	2	2	0	1	11
MR043	MoleRec	2	1	2	2	2	2	1	12
MR044	AKA-SafeMed	2	2	2	2	2	1	1	12
MR045	SSPNet	2	2	2	2	2	1	1	12
MR046	HKRec	2	2	2	2	2	1	2	13
MR047	KEDRec-LM	2	2	1	1	0	1	1	8
MR048	MedRec	2	2	2	2	0	2	2	12

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Table R2 continued

ID	Study/model	C1	C2	C3	C4	C5	C6	C7	Total
MR049	KEHGCN	2	2	2	2	2	2	2	14
MR050	BH3-MedRec	2	2	2	2	2	2	2	14
MR051	LAMO	2	2	2	2	1	1	2	12
MR052	RAREMed	2	2	2	2	2	0	2	12
MR053	MKAF-Net	2	2	2	2	1	1	2	12
MR054	TCM-KLLaMA	2	2	2	2	1	1	1	11
MR055	DAI-Net	2	1	2	2	2	2	2	13
MR056	RPNet	2	2	2	2	2	1	2	13
MR057	Nephrotoxic medication RL	2	1	2	2	2	2	2	13
MR058	MetaDrug	2	2	2	2	2	0	2	12
MR059	HypeMed	2	2	2	2	2	1	1	12
MR060	GenRxR	2	2	2	2	2	1	2	13

The 13 newly scored studies are REFINE, context-aware MR (Carmen), HI-DR, IMDR, BiMoRec, SDRBT, HKRec, LAMO, RPNet, nephrotoxic medication RL, MetaDrug, HypeMed, and GenRxR. The evidence anchors and rationales for C5–C7 are in `appraisal_additions_13.csv`. For example, IMDR’s missing-view stress tests support C7 while its visualizations support only partial C6; the RL paper’s inconsistent cohort totals yield partial C2; and GenRxR’s expert test concerns synthetic-record plausibility, not the safety of predicted regimens. These ratings do not alter any underlying study-selection counts.

Criterion	Adequate (2)	Partial (1)	Inadequate/NR (0)
C1	59	1	0
C2	53	7	0
C3	57	3	0
C4	58	2	0
C5	46	8	6
C6	20	33	7
C7	36	23	1

R5. Endpoint-level extraction for the five additions

These findings are based on inspected full texts and must not be extrapolated to the full 60-study register. “Not reported” refers to the inspected source. Four of the five additions report DDI rate; the RL study instead evaluates adverse-outcome discrimination. None of the five establishes prospective clinical harm reduction. Only GenRxR reports a licensed expert’s blinded assessment, concerning synthetic data.

- **RPNet:** clinical-code matching pretraining and rare-code-weighted similar-visit retrieval; binary cross-entropy, multilabel, and DDI objectives. MIMIC-III/IV F_1 : 0.7004/0.6231; DDI rate: 0.0737/0.0723. Test-code deletion evaluates noise sensitivity. Case interpretation does not constitute expert clinical validation; probability calibration, explicit contraindication outcomes, and prospective safety are not reported. Source: publisher PDF pp.6–10, Tables 2–3.
- **Nephrotoxic medication RL:** nine medication actions, offline RL with dead-end discovery, and integrated-gradient feature attributions. Endpoints include mortality, stage-3 AKI, renal replacement therapy, and septic shock. ICU-mortality IQL AUROC: 0.852 internally, 0.845 in a later same-hospital cohort. This is discrimination and temporal validation, not deployed-policy outcome benefit or external institutional validation. No formal probability-calibration or DDI-rate endpoint is identified. Source: Sections 2–4 and Appendix A.

- **MetaDrug:** patient/visit Transformers, self/peer patient adaptation, and uncertainty filtering of support visits. MIMIC-III and Kansas AKI are trained/evaluated within each dataset. DDI rate and code-diversity subgroups are evaluated, but the support-set filter is not output probability calibration. Clinical safety, hard contraindication evaluation, and locked-model institutional transfer are not established. Source: arXiv:2601.22820v1, Sections II–III and Tables III–VI.
- **HypeMed:** knowledge-aware hypergraph pretraining and similar-visit retrieval with longitudinal fusion. Three retrospective benchmarks are evaluated separately. DDI loss/rate apply to MIMIC-III/IV; eICU omits them because the DDI graph is unavailable. The multicenter eICU benchmark therefore adds predictive evidence without a corresponding safety endpoint. No expert recommendation-safety assessment or probability-calibration result is reported. Source: inspected 28-page PDF, Sections 5.1–5.9.
- **GenRxR:** GPT-4o creates counterfactual records; LLaMA 3.1 8B Instruct is adapted for recommendation. DDI and rare/frequent medication-group metrics are evaluated. One licensed expert distinguishes 50 original from 50 synthetic records in a blinded plausibility test. This does not assess clinical safety of recommended regimens, systematic rationale faithfulness, or causal treatment effects. Drug–disease conflict screening is listed as future work. Source: inspected 9-page PDF, Sections 5.1–5.5.

`revision_evidence_extraction.csv` separates DDI, contraindication, calibration, robustness, grounding/hallucination, validation, and expert evidence. Full-corpus endpoint frequencies cannot be recovered from the ordinal appraisal matrix and require a new extraction pass.

R6. Source uncertainties and work required before resubmission

The submitted and later manuscripts report different screening-stage allocations despite the same original 56-study total. The earlier submitted version reports 32 title/abstract exclusions, 61 full texts and five full-text exclusions; the supplied updated source reports 37 title/abstract exclusions, 56 full texts and no original full-text exclusions. Original record-level logs are required to reconcile this history; the revision diagram preserves the updated source’s history and explicitly separates the new scope correction.

Author records are also needed to confirm which eligibility, extraction, taxonomy, and appraisal criteria were prespecified, the independent appraisal-reviewer process, disagreement resolution, and complete database execution details/PRISMA checklist. The original screening-reviewer description is not new evidence of independent appraisal.

The uploaded RL paper has inconsistent cohort totals between Section 2.1 and Appendix Table A.5 and some inconsistent confidence-interval entries. Cohort totals and problematic intervals are therefore not reproduced. GenRxR’s DDI values require table/prose consistency checking before numerical quotation. FLAME names different Aloe checkpoints in the main text and Appendix C; the illustrative LLM table flags this. The 13 additional appraisal ratings require author cross-check; full-corpus endpoint counts and a complete LLM/RAG evidence table remain outstanding. The original S1 search documentation should be synchronized with this addendum once the authors confirm the historical logs.

R7. Initial six-paper full-text endpoint verification

All six papers below already occur in the 60-study register. Their full-text verification does not add studies or change the PRISMA identification/screening counts. This initial selected evidence pass is not a representative sample or a full-corpus frequency estimate; Section R8 reports the subsequent, broader endpoint recheck.

E = empirical endpoint reported; P = qualitative or component-level proxy only; NR = no direct endpoint identified in the supplied main full text. Supplementary files referenced by the original articles

were not supplied for this pass. U = awaiting endpoint verification. U must not be merged with NR. Robustness includes an explicit perturbation or distribution-shift test, rather than an architectural claim or ordinary ablation. Grounding/hallucination here requires evaluation of recommendation support or factual errors; generic clinical-quality ratings and upstream retrieval tests are retained as proxies. External validation means a model developed on one institutional source is tested on another source without target-cohort fine-tuning. These extraction rules are documented for this revision pass and are not claimed to have been prespecified before the original search.

Study	DDI	Contra.	Calib.	Robust.	Grounding	External
Confidence calibration	NR	NR	E	E	NR	NR
KRAM	P	NR	NR	E	NR	NR
PAUP	E	NR	NR	NR	NR	NR
KATMed	NR	E	NR	E	P	E
RAG-CPMF	NR	NR	NR	P	P	NR
TCM-KLLaMA	NR	NR	NR	P	P	NR

All six report quantitative task/performance evaluation. Only RAG-CPMF reports a dedicated expert recommendation-quality assessment in these six main texts: three TCM clinicians assess 102 standardized cases. KATMed’s knowledge-extraction plausibility checks do not establish expert evaluation of predicted medication sets. This distinction is also retained in the endpoint register.

- **Confidence calibration:** ECE, NLL and ACC^+ assess individual confidence; MSE/ECE assess medication-set estimates. Label-disturbance experiments and calibration-set-size sensitivity are reported. Calibration is agreement with historical prescription labels, not validation of treatment safety. Source: Sections 5.1–5.3, Tables 1–3 and Appendix Tables 4–5 (supplied PDF pp.5–12).
- **KRAM:** controlled label replacement/addition at noise ratios 0.1–0.4; aggregate predictive metrics in Tables 6–7. DDIs appear in a one-patient case study (Table 9), without an aggregate DDI rate. No output-probability calibration or external cohort validation is identified. Source: Sections 5.1–5.5, supplied PDF pp.8–17.
- **PAUP:** separate MIMIC-III/IV benchmarks report DDI rate and prescription overlap. Hyperparameter sensitivity, ablations, patient-state variants and runtime analyses do not establish controlled corruption/subgroup robustness or external model transfer. Source: Sections 4.1–4.7, Tables 1–6, supplied PDF pp.8–13.
- **KATMed:** evaluates DDCC-VR and reports MIMIC-III-to-IV testing and fixed-model eICU testing without fine-tuning (Tables 9–10). The main text gives conflicting split ratios (Section 6.1: 4:1:1; Section 7.1: 7:1:2) and inconsistent descriptions of manual review during knowledge extraction. These concerns qualify reproducibility and expert-validation claims; reported external metrics remain retrospective. Source: Sections 4.2, 6.1–7.1 and 8.7, supplied PDF pp.6,9–15.
- **RAG-CPMF:** Qwen2.5–72B–Instruct retrieves the top three semantically matched CPM instructions. A public instruction dataset supports 102 guideline-derived cases spanning 34 diseases; correctness accepts any top-three guideline match. Three clinicians grade safety/usability/smoothness. These are clinical-quality proxies, without a separate hallucination, evidence-citation faithfulness, DDI or contraindication-violation endpoint. Source: Sections 2.4–2.5, 3.4 and 4.3–4.4, Figure 9, supplied PDF pp.6,9–12.
- **TCM-KLLaMA:** Chinese-LLaMA2-7B with LoRA and an 811-herb classification head; public Dictionary of TCM Prescriptions benchmark, 33,765 cleaned pairs, random 8:1:1 split. Synonym-recognition accuracy is a component-level test. Vocabulary constraints and overlap ablations do not directly evaluate hallucination or rationale faithfulness. DDI integration and clinical validation are future work. Source: Sections 4.1–4.5 and 5, Tables 1–3, supplied PDF pp.5–8.

Appraisal corrections. KRAM changes from (2,1,1,1,0,0,2), total 7, to (2,2,2,2,1,1,2), total 12: dataset/preprocessing, architecture/training and predictive evaluation are documented; the DDI and graph case study gives partial safety/explanation evidence. KATMed C2 changes 2 to 1 (total 13 to 12) because conflicting split descriptions prevent an unqualified adequate-transparency rating. TCM-KLLaMA C2 changes 1 to 2 (total 10 to 11) because the public source, cleaning counts, vocabulary and split are described. Other scores for the six checked studies are retained. These are single-pass descriptive corrections, not an independent multi-reviewer risk-of-bias assessment. The remaining 41 original ratings are inherited; the 13 previously unrated studies have now been scored in R4, pending author cross-check. The original values remain in `appraisal_subset_47_original.csv`.

Audit continuation. `safety_endpoint_audit_60.csv` contains all 60 stable IDs. The initial six-study check was extended through accessible original articles and publisher descriptions; the current state and remaining uncertainties are documented in Section R8. Full counts for Reviewer 1 Comment 4 and Reviewer 2’s corresponding safety-evidence comment remain outstanding while endpoint-specific U entries remain.

R8. Extended endpoint recheck (4 October 2026)

The original S1 recorded study identities, broad model families, datasets, and seven ordinal quality criteria. It did not preserve a per-study column for each of the six safety and reliability endpoints requested by the reviewers. The current manuscript names many relevant examples; these examples cannot themselves be summed to obtain complete 60-study endpoint frequencies. The issue was a missing *endpoint-specific recoding*, not a claim that all 54 previously unrecoded studies lacked extracted information.

The machine-readable `safety_endpoint_audit_60.csv` now includes source anchors and study-level classifications; this appraisal revision additionally verified IMDR missing-view robustness and HKRec DDI and visit-count subgroup results from the uploaded full texts. In addition to the original six full-text checks, 52 other studies have at least one endpoint checked against an available original article, author manuscript, indexed primary publisher description, or authors’ implementation. Two studies still have no endpoint-specific code; other studies remain U for individual dimensions not yet checked. This is an incremental evidence audit, not a completed independent double extraction. For publisher-abstract-only checks, negative findings remain U rather than NR.

Evaluation dimension	E	P	NR	U
Predictive performance	58	0	0	2
DDI risk (including nonstandard DDI measures)	33	7	14	6
Contraindication violations	1	2	5	52
Probability calibration	1	0	5	54
Explicit robustness or subgroup stress test	14	2	1	43
Recommendation grounding or hallucination	1	6	3	50
Fixed-model external institutional validation	3	0	6	51

The E column is a verified *minimum* within the 60-study register, not a final prevalence estimate. A U is never treated as unreported. DDI E includes REFINE’s severity-weighted rate, the nonstandard adverse-DDI count of context-aware MR (Chen et al.), and ordinary aggregate DDI rate; these should be disaggregated when publishing final results. The grounding E of TreatRAG tests invalid ATC-3 medication terms, not source citation accuracy or rationale faithfulness. KATMed is the confirmed empirical contraindication-violation example; KEHGCN has a contraindication penalty but no separate violation-rate result in its inspected paper. MedGCN and MedRec are confirmed examples with predictive evaluation but no DDI-rate endpoint. MedRec evaluates TCM and private symptom-set data, rather than the MIMIC-III dataset formerly listed in S1.

GPSRec and MR-DTR remain entirely endpoint-unclassified in the current matrix. Original publisher access was insufficient to code their metrics from primary sources. Supplying their result tables

or complete PDFs would resolve these rows. The DDI result tables of Trans-GAHNet, TAHNet, AKA-SafeMed, and DAI-Net also need checking: their accessible abstracts, claims, or authors' code do not establish a paper-level empirical DDI result. Many dimension-specific U cells across the other 58 studies, especially contraindication, calibration, and rationale-grounding assessment, also require a systematic full-text negative check. The 60-study C1–C7 appraisal is not an endpoint matrix; historical ratings were not recalculated from this partial endpoint audit. Source metadata corrections in R3 do not, by themselves, change a paper's appraisal score.

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